9. 5 Absolute Convergence; The Ratio and Root Tests

Definition 1:

(1) If $\sum|a_n|$ converges, we say $\sum a_n$ ***converges absolutely***.

(2) If $\sum a_n$ converges but $\sum|a_n|$ diverges, we say $\sum a_n$ ***converges conditionally.***

(Section 9.6)

Example 1:

(1) $\sum_{n=1}^{\infty}\left(-\frac{1}{e}\right)^{n-1}$ converges absolutely since $\sum_{n=1}^{\infty}\left|\left(-\frac{1}{e}\right)^{n-1}\right| = \sum_{n=1}^{\infty}\left(\frac{1}{e}\right)^{n-1}$ converges.

(2) $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}$ converges absolutely since $\sum_{n=1}^{\infty}\left|\frac{(-1)^{n+1}}{n^2}\right| = \sum_{n=1}^{\infty}\frac{1}{n^2}$ converges.

(3) $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}$ converges conditionally since $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}$ converges but $\sum_{n=1}^{\infty}\left|\frac{(-1)^{n+1}}{n}\right| = \sum_{n=1}^{\infty}\frac{1}{n}$ diverges.

Theorem 1 (The Absolute Convergence Test):

$\sum|a_n|$ converges $\Rightarrow \sum a_n$ converges.

Proof:

For each n, $\quad -|a_n| \le a_n \le |a_n| \;\Rightarrow\; 0 \le a_n + |a_n| \le 2|a_n|$.

If $\sum|a_n|$ converges, then $\sum 2|a_n|$ converges and, by the Direct Comparison Test, the nonnegative series $\sum(a_n + |a_n|)$ converges. Then,

$$\sum a_n = \sum(a_n + |a_n| - |a_n|) = \sum(a_n + |a_n|) - \sum|a_n|.$$

Therefore, $\sum a_n$ converges since both $\sum(a_n + |a_n|)$ and $\sum|a_n|$ converge.

Take Note: The converse of Theorem 1 is not true.

When $\sum|a_n|$ diverges, $\sum a_n$ may converge or may diverge.

Example 2:

1. $\sum_{n=1}^{\infty}\left|\frac{(-1)^{n+1}}{n}\right| = \sum_{n=1}^{\infty}\frac{1}{n}$ diverges but $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}$ converges.

2. $\sum_{n=1}^{\infty}|(-2)^{n+1}| = \sum_{n=1}^{\infty}2^{n+1}$ and $\sum_{n=1}^{\infty}(-2)^{n+1}$ both diverge.

3. $\sum_{n=1}^{\infty}\frac{\sin n}{n^2}=\frac{\sin 1}{1}+\frac{\sin 2}{4}+\frac{\sin 3}{9}+\frac{\sin 4}{16}+\frac{\sin 5}{25}+\cdots$ which contains both positive and negative terms. In this case, we test for the series $\sum_{n=1}^{\infty}\left|\frac{\sin n}{n^2}\right|$.

$\forall n\geq 1,\ 0\leq\left|\frac{\sin n}{n^2}\right|\leq\frac{1}{n^2}$ and $\sum_{n=1}^{\infty}\frac{1}{n^2}$ is a convergent *p*-series.

By the direct comparison test, $\sum_{n=1}^{\infty}\left|\frac{\sin n}{n^2}\right|$ converges.

Hence, $\sum_{n=1}^{\infty}\frac{\sin n}{n^2}$ converges by the absolute convergence test.

The Ratio Test

The Ratio Test measures the rate of growth (or decline) of a series by examining the ratio $\frac{a_{n+1}}{a_n}$.

For a geometric series $\sum ar^n$, $\frac{a_{n+1}}{a_n}=r$ is a constant, and the series converges if and only if its ratio is less than 1 in absolute value. The Ratio Test is a powerful rule extending that result.

Theorem 2 (The Ratio Test):

Let $\sum a_n$ be any series and suppose that $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\rho$.

Then (a) the series converges absolutely if $\rho<1$,

(b) the series diverges if $\rho>1$ or is infinite,

(c) the test is inconclusive if $\rho=1$.

Remark:

1. **Do not apply the Ratio Test when $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=1$ or limit does not exist (except when limit is ∞).**

Example 1: Let $a_n=\frac{1}{n}$ and $b_n=\frac{1}{n^2}$

Then $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\left|\frac{1}{n+1}(n)\right|=1$ and $\lim_{n\to\infty}\left|\frac{b_{n+1}}{b_n}\right|=\lim_{n\to\infty}\left|\frac{1}{(n+1)^2}\left(n^2\right)\right|=1$.

In both cases, $\rho=1$, yet $\sum_{n=1}^{\infty}\frac{1}{n}$ diverges, whereas $\sum_{n=1}^{\infty}\frac{1}{n^2}$ converges.

2. The Ratio Test **often effective** when the terms of a series contain **factorials of expression involving *n* or expressions raised to a power involving *n***. The test is **NOT suitable for the series with a_n is rational or algebraic** because $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=1$.

Example 3: Investigate the convergence of the following series.

(1) $\sum_{n=1}^{\infty}\frac{2^n+5}{3^n}$ (2) $\sum_{n=1}^{\infty}\frac{(2n)!}{n!n!}$ (3) $\sum_{n=1}^{\infty}\frac{4^n\,n!n!}{(2n)!}$

Solution:

(1) $$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\left|\frac{2^{n+1}+5}{3^{n+1}}\left(\frac{3^n}{2^n+5}\right)\right|=\lim_{n\to\infty}\frac{1}{3}\left(\frac{2^{n+1}+5}{2^n+5}\right)=\frac{1}{3}\lim_{n\to\infty}\left(\frac{2+\frac{5}{2^n}}{1+\frac{5}{2^n}}\right)=\frac{2}{3}<1.$$

By the Ratio Test, $\sum_{n=1}^{\infty}\frac{2^n+5}{3^n}$ converges absolutely.

(2) $$\lim_{n\to\infty}\frac{a_{n+1}}{a_n}=\lim_{n\to\infty}\frac{(2(n+1))!}{(n+1)!(n+1)!}\left(\frac{n!\,n!}{(2n)!}\right)=\lim_{n\to\infty}\frac{(2n+2)(2n+1)(2n)!}{(n+1)(n!)(n+1)(n!)}\left(\frac{n!\,n!}{(2n)!}\right)$$

$$=\lim_{n\to\infty}\frac{2(n+1)(2n+1)}{(n+1)(n+1)}=\lim_{n\to\infty}\frac{4n+2}{n+1}=\lim_{n\to\infty}\frac{4+\frac{2}{n}}{1+\frac{1}{n}}=4>1.$$

By the Ratio Test, $\sum_{n=1}^{\infty}\frac{(2n)!}{n!n!}$ diverges.

(c) $$\lim_{n\to\infty}\frac{a_{n+1}}{a_n}=\lim_{n\to\infty}\frac{4^{n+1}(n+1)!(n+1)!}{(2(n+1))!}\left(\frac{(2n)!}{4^n\,n!n!}\right)=\lim_{n\to\infty}\frac{4(n+1)(n+1)}{(2n+2)(2n+1)}$$

$$=\lim_{n\to\infty}\frac{2(n+1)}{2n+1}=\lim_{n\to\infty}\frac{2+\frac{2}{n}}{2+\frac{1}{n}}=1.$$

We **<u>cannot decide from the Ratio Test</u>** whether $\sum_{n=1}^{\infty}\frac{4^n\,n!n!}{(2n)!}$ converges.

Note that $\frac{a_{n+1}}{a_n}=\frac{2n+2}{2n+1}>1 \Rightarrow a_{n+1}>a_n \quad n=1,2,3,\cdots$

$\Rightarrow a_n > a_1=2 \ \forall\ n=2,3,4,\cdots \Rightarrow \lim_{n\to\infty}a_n>2.$

$\therefore \lim_{n\to\infty}a_n\neq 0$. By the *n*-th term test for divergence, the series diverges.

The Root Test

Theorem 3 (The Root Test):

Let $\sum a_n$ be any series and suppose that $\lim_{n\to\infty} \sqrt[n]{|a_n|} = \rho$.

Then (a) the series converges absolutely if $\rho < 1$,

(b) the series diverges if $\rho > 1$ or is infinite,

(c) the test is inconclusive if $\rho = 1$.

Remark:

1. **Do not apply the Root Test when $\lim_{n\to\infty} \sqrt[n]{|a_n|} = 1$ or limit does not exist (except when limit is ∞).**

   Example 4: Let $a_n = \frac{1}{n}$ and $b_n = \frac{1}{n^2}$

   Then $\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{\left|\frac{1}{n}\right|} = \lim_{n\to\infty} \frac{1}{\sqrt[n]{n}} = 1$ and $\lim_{n\to\infty} \sqrt[n]{|b_n|} = \lim_{n\to\infty} \sqrt[n]{\left|\frac{1}{n^2}\right|} = \lim_{n\to\infty} \frac{1}{\left(\sqrt[n]{n}\right)^2} = 1.$

   In both cases, $\rho = 1$, yet $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, whereas $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

2. Suitable when a_n is of the form $\left(b_n\right)^n$.

3. $\lim_{n\to\infty} \sqrt[n]{n} = \lim_{n\to\infty} n^{1/n} = 1$ is useful.

Example 5: Investigate the convergence of the following series.

$$\frac{1}{2} + \frac{1}{4} + \frac{3}{8} + \frac{1}{16} + \frac{5}{32} + \frac{1}{64} + \frac{7}{128} + \cdots$$

Solution:

The n-th term of the series is $a_n = \begin{cases} n/2^n, & n \text{ odd} \\ 1/2^n, & n \text{ even} \end{cases}$.

Then, $\sqrt[n]{|a_n|} = \begin{cases} \sqrt[n]{n}/2, & n \text{ odd} \\ 1/2, & n \text{ even} \end{cases}$. Hence, $\frac{1}{2} \le \sqrt[n]{|a_n|} \le \frac{\sqrt[n]{n}}{2} \quad \forall\, n \ge 1.$

Since $\lim_{n\to\infty} \frac{\sqrt[n]{n}}{2} = \frac{1}{2} = \lim_{n\to\infty} \frac{1}{2}$, by the Sandwich Theorem, $\lim_{n\to\infty} \sqrt[n]{|a_n|} = \frac{1}{2} < 1.$

By the Ratio Test, the series converges absolutely.

Example 6: Which of the following series converge, and which diverge?

1. $\sum_{n=1}^{\infty} \frac{n^5}{5^n}$
2. $\sum_{n=1}^{\infty} \frac{2^n}{n^3}$
3. $\sum_{n=2}^{\infty} \frac{n^n}{2^{(n^2)}}$
4. $\sum_{n=2}^{\infty} \left(\frac{1}{1+n}\right)^n$
5. $\sum_{n=2}^{\infty} \frac{n}{(\ln n)^n}$

Solution:
We apply the Root Test to each series, noting that each series has positive terms.

1. $\lim_{n\to\infty} \sqrt[n]{a_n} = \lim_{n\to\infty} \sqrt[n]{\frac{n^5}{5^n}} = \lim_{n\to\infty} \frac{\left(\sqrt[n]{n}\right)^5}{\left(\sqrt[n]{5}\right)^n} = \frac{\lim_{n\to\infty}\left(\sqrt[n]{n}\right)^5}{5} = \frac{1^5}{5} < 1. \left(\because \lim_{n\to\infty} \sqrt[n]{n} = 1.\right)$

   By the Root Test, $\sum_{n=1}^{\infty} \frac{n^5}{5^n}$ converges.

   Alternative method: Use the RatioTest

   $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(n+1)^5}{5^{n+1}}\left(\frac{5^n}{n^5}\right)\right| = \lim_{n\to\infty} \frac{1}{5}\left(\frac{n+1}{n}\right)^5 = \frac{1}{5}\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^5 = \frac{1}{5} < 1.$

   By the Ratio Test, $\sum_{n=1}^{\infty} \frac{n^5}{5^n}$ converges.

2. $\lim_{n\to\infty} \sqrt[n]{a_n} = \lim_{n\to\infty} \sqrt[n]{\frac{2^n}{n^3}} = \lim_{n\to\infty} \frac{\left(\sqrt[n]{2}\right)^n}{\left(\sqrt[n]{n}\right)^3} = \frac{2}{\lim_{n\to\infty}\left(\sqrt[n]{n}\right)^3} = \frac{2}{1^3} > 1. \left(\because \lim_{n\to\infty} \sqrt[n]{n} = 1.\right)$

   By the Root Test, $\sum_{n=1}^{\infty} \frac{2^n}{n^3}$ converges.

   Alternative method: Use the Ratio Test

   $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{2^{n+1}}{(n+1)^3}\left(\frac{n^3}{2}\right)\right| = \lim_{n\to\infty} 2\left(\frac{n}{n+1}\right)^3 = 2\lim_{n\to\infty}\left(\frac{1}{1+(1/n)}\right)^3 = 2 > 1.$

   By the Ratio Test, $\sum_{n=1}^{\infty} \frac{2^n}{n^3}$ diverges.

3. $\lim\limits_{n\to\infty}\sqrt[n]{a_n}=\lim\limits_{n\to\infty}\sqrt[n]{\dfrac{n^n}{(2^n)^n}}=\lim\limits_{n\to\infty}\dfrac{n}{2^n}$.

Let $f(x)=\dfrac{x}{2^x}$, $x\geq 1$. Then the function agrees with $f(n)=\dfrac{n}{2^n}$ at positive integers.

$\therefore\ \lim\limits_{n\to\infty}\sqrt[n]{|a_n|}=\lim\limits_{n\to\infty}\dfrac{n}{2^n}=\lim\limits_{n\to\infty}\dfrac{x}{2^x}=\lim\limits_{n\to\infty}\dfrac{1}{2^x\ln 2}=0<1.$

By the Root Test, $\sum\limits_{n=2}^{\infty}\dfrac{n^n}{2^{(n^2)}}$ converges.

4. $\lim\limits_{n\to\infty}\sqrt[n]{a_n}=\sqrt[n]{\left(\dfrac{1}{1+n}\right)^n}=\lim\limits_{n\to\infty}\dfrac{1}{1+n}=0<1.\ \left(\because \lim\limits_{n\to\infty}\sqrt[n]{n}=1.\right)$

By the Root Test, $\sum\limits_{n=2}^{\infty}\left(\dfrac{1}{1+n}\right)^n$ converges.

5. $\lim\limits_{n\to\infty}\sqrt[n]{a_n}=\sqrt[n]{\dfrac{n}{(\ln n)^n}}=\lim\limits_{n\to\infty}\dfrac{\sqrt[n]{n}}{\ln n}=0<1.\ \left(\because \lim\limits_{n\to\infty}\sqrt[n]{n}=1.\right)$

By the Root Test, $\sum\limits_{n=2}^{\infty}\dfrac{n}{(\ln n)^n}$ converges.